

JACK T. DINSMORE

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RESEARCH INTERESTS

- Compact objects: their direct emission and the surrounding winds, outflows, and nebulae
- Advancing analysis techniques for X-ray and optical astronomical data, including polarization
- Cosmic rays and magnetic fields in the interstellar medium

EDUCATION

STANFORD UNIVERSITY 2022 – 2027 (projected)

Ph.D. in Physics (GPA: 4.0/4.0)

ADVISOR: Prof. Roger W. Romani

MASSACHUSETTS INSTITUTE OF TECHNOLOGY 2018 – 2022

B.S. in Physics, Minors in Astronomy and Mathematics, Concentration in Music (GPA: 5.0/5.0)

RESEARCH MENTORS: Profs. Tracy R. Slatyer, Julien de Wit, and Philip Harris

AWARDS & HONORS

- NSF Graduate Research Fellowship Program Honorable Mention 2024
- Inducted into ΦBK and $\Sigma\Pi\Sigma$ honors societies 2022
- MIT Barrett Prize for excellence in astrophysics research 2022
- Lehigh University REU program 2020

TELESCOPE ALLOCATIONS

OBSERVING RECORD: 11 successful grant proposals (10 as co-I, 1 as PI). >2 Ms X-ray observatory time, ~8 nights optical observing experience.

- *NuSTAR*: 500 kiloseconds (1 grant as co-I)
- Imaging X-ray Polarimetry Explorer: 1 megasecond (1 grant as co-I)
- *Chandra* X-ray observatory, ACIS: 510 kiloseconds (3 grants as co-I)
- Magellan Observatory, proto-Lightspeed: 6 nights (3 grants as co-I)
- National Radio Astronomy Observatory: 50 minutes (1 grant as co-I)

ADDITIONAL GRANT ALLOCATIONS

- Kavli Institute: KICK grant for developing new technology (1 grant as PI)
- KIPAC: Innovation grant (1 grant as co-I)

RESEARCH MENTORING

- Stanford graduate student Yifan Hu 2026
- Stanford undergraduate student Calvin McCullumsmith 2026
- Stanford undergraduate student Tristan Jau-Shi Lu 2026
- Stanford graduate student Ningyue (Fanny) Fan 2026
- Stanford undergraduate student Peter Lindholm 2025
- Stanford graduate student Kaitlyn Karpovich 2024

TEACHING

- Mentor, Stanford directed reading program 2025
- TA, Stanford PHYSICS 130: Quantum Mechanics (rated 4.41/5 in effectiveness) Spring 2025
- TA, Stanford PHYSICS 110: Advanced Mechanics (rated 4.36/5 in effectiveness) Fall 2024
- TA, Stanford PHYSICS 120: Intermediate E&M (rated 4.73/5 in effectiveness) Winter 2024
- TA, Stanford PHYSICS 44: Lab for E&M (rated 4.71/5 in effectiveness) Spring 2023

- Course material designer and TA, MIT 8.S50: Statistics Winter 2022
- Grader, MIT 8.012: Physics I Spring 2019
- Teacher, MIT Academic Teaching Initiative 2018

PROFESSIONAL SERVICE & OUTREACH

- Reviewer for the Astrophysical Journal 2026
- Member, KIPAC communications director search committee 2026
- Member, KIPAC Tea committee 2025–2026
- Editor, Writer, and Web Manager for the KIPAC Research Highlights 2023–2026
- Co-organizer, KIPAC computing bootcamp 2022
- Volunteer, KIPAC stargazing 2025
- Volunteer, KIPAC community day (science fair) 2025
- Volunteer, Stempalooza (science fair) 2024
- Mentor, Stanford Future Advancers of Science and Technology 2023–2024
- Organizer, KIPAC community day (science fair) 2023

PRESS COVERAGE

- NASA press release “[NASA Space Telescope Maps Magnetic Fields of ‘Lighthouse’ Pulsar](#)” 2026
- “[KIPAC Year in Review](#)” 2024–2025
- “[KIPAC Years in Review](#)” 2022–2024
- MIT News: “[Method for decoding asteroid interiors could help aim asteroid-deflecting missions](#)” 2022
- Astrobit: “[Using tides to peak into asteroid interiors](#)” 2022

INVITED PRESENTATIONS

- “IXPE Polarizations of the Lighthouse X-ray Filament,” MAT talk (MIT) 2026
- “Untangling Pulsar Magnetic Fields with Optical and X-ray Polarizations,” THEA talk (Columbia University) 2026
- “Mysterious Pulsar X-ray Filaments: Observational and Theoretical Advances,” Tea talk (Caltech) 2026
- “Untangling Pulsar Magnetic Fields with Optical and X-ray Polarizations,” Carnegie Observatories 2026
- “proto-Lightspeed: an MKI-KIPAC Project to Capture the Changing Polarized Sky,” Kavli Symposium (University of Cambridge) 2026
- “proto-Lightspeed: a Low Noise, High Speed Imager for Magellan,” KIPAC Tea (Stanford) 2026
- “IXPE Polarimetry of Pulsar Wind Nebulae,” Friends of IXPE meeting (remote) 2026

CONTRIBUTED PRESENTATIONS

- American Astronomical Society Summer meeting (Pasadena, CA) 2026
- IXPE PWN group (remote) 2026
- Compact Objects Group, (Stanford) 2026
- Massachusetts Institute of Technology CMOS group (MIT) 2025
- Imaging X-ray Polarimetry Explorer SASWG group (remote) 2025
- International X-ray Polarimetry Symposium (Huntsville, AL) 2024
- Fields, Flows, and Filaments in the Magnetic ISM (Stanford) 2024
- Asteroids, Comets, and Meteorites conference (Flagstaff, AZ) 2023
- Imaging X-ray Polarimetry Explorer SASWG group (remote) 2023
- Apophis T–7 Years (remote) 2022

LEAD AUTHOR REFEREED PUBLICATIONS

PUBLICATION RECORD: 24 research articles, 3 public software libraries. Cited in ~ 220 academic works. h -index of 7.

0. **Jack T. Dinsmore**, Christopher Layden, Roger W. Romani, Vera L. Berger, Kevin B. Burdge, and others. **Optimal Photon Counting with Fast, Low Noise Astronomical Imagers**. *AJ* (submitted)
- 1. **Jack T. Dinsmore**. **A Statistical Mechanics Model for Stable Rings Outside the Roche Limit**. *Icarus*, 461:117302, 2027
- 2. **Jack T. Dinsmore**, Roger W. Romani, S. Zhang, C.-Y. Ng, Stefano Silvestri, and others. **IXPE Polarizations of the Lighthouse Pulsar, Trail, and Filament**. *ApJ*, 1005(2):237, July 2026
- 3. **Jack T. Dinsmore** and Roger W. Romani. **Chandra Proper Motions and Milliarcsecond Astrometry of Nineteen Pulsars**. *ApJ*, 1000(2):183, March 2026
- 4. **Jack T. Dinsmore** and Roger W. Romani. **Advanced Weights for IXPE Polarization Analysis**. *ApJ*, 993(2):173, November 2025
- 5. **Jack T. Dinsmore**, Roger W. Romani, Nikos Mandarakas, Dmitry Blinov, and Ioannis Liodakis. **The Guitar Filament's Magnetic Field Revealed by Starlight Polarization**. *ApJ*, 980(2):229, February 2025
- 6. **Jack T. Dinsmore** and Roger W. Romani. **A Catalog of Pulsar X-Ray Filaments**. *ApJ*, 976(1):4, November 2024
- 7. **Jack T. Dinsmore** and Roger W. Romani. **Polarization Leakage and the IXPE Point-Spread Function**. *ApJ*, 962(2):183, February 2024
- 8. **Jack T. Dinsmore** and Julien de Wit. **Constraining the Interiors of Asteroids Through Close Encounters**. *MNRAS*, 520(3):3459–3475, 10 2022
- 9. **Jack T. Dinsmore** and Tracy R. Slatyer. **Luminosity Functions Consistent with a Pulsar-Dominated Galactic Center Excess**. *JCAP*, 06(06):025, 2022

LEAD AUTHOR NON-REFEREED PUBLICATIONS AND SOFTWARE

0. **Jack T. Dinsmore**. **jtdinsmore/lightspeedpy: Data Reduction Tools for Magellan/Proto-Lightspeed**., September 2026
- 1. **Jack T. Dinsmore** and Roger W. Romani. **A Physical Model of Pulsar X-Ray Filaments**. *arXiv-prints*, page 2603.20532, March 2026
- 2. **Jack T. Dinsmore**. **jtdinsmore/leakagelib: Advanced Weights for IXPE Polarimetry**., September 2025
- 3. **Jack T. Dinsmore**. **jtdinsmore/ixpenn: IXPE Neural Net Event Reconstruction Pipeline**., September 2025

CONTRIBUTING AUTHOR REFEREED PUBLICATIONS

0. Philip Kaaret, **Jack T. Dinsmore**, Naomi Tsuji, Samar Safi-Harb, Ningyue Fan, and others. **X-Ray Polarization of the Western Lobe of SS 433**. *ApJ* (submitted)
- 1. Ningyue Fan, **Jack T. Dinsmore**, and Roger W. Romani. **SS 433 Eastern Jet Turbulence Revealed by Spatially-Resolved IXPE Data Analysis**. *ApJ* (submitted)
- 2. Josephine Wong, **Jack T. Dinsmore**, Roger W. Romani, Stefano Silvestri, Shumeng Zhang, and others. **Kes 75 with IXPE: Detection of Nebular X-Ray Polarization and Change in Pulsar Lightcurve**. *ApJ*, 1006(2):113, August 2026
- 3. Christopher Layden, Kevin Burdge, Gabor Furesz, Juliana García-Mejía, **Jack T. Dinsmore**, and others. **Proto-Lightspeed: A High-Speed, Ultra-Low Read Noise Imager on the Magellan Clay Telescope**. *PASP*, 138(6):065001, June 2026
- 4. Paolo Soffitta, Niccolò Bucciantini, Josephine Wong, Denis González-Caniulef, Matteo Bachetti, and others. **IXPE View of the Crab Pulsar Following the 2025 July 17 and August 6 Glitches**. *ApJ*, 1001(2):219, April 2026
- 5. Emma T. Chickles, Joheen Chakraborty, Kevin B. Burdge, Vik S. Dhillon, Paul Draghis, and others. **An Eclipsing 8.56 Minute Orbital Period Mass-Transferring Binary**. *ApJ*, 1000(2):237, March 2026
- 6. Andrew G. Sullivan, **Jack T. Dinsmore**, and Roger W. Romani. **X-Ray Polarization of the Intrabinary Shock in Redback Pulsar J1723–2837**. *ApJ*, 999(1):128, March 2026
- 7. Niccolò Di Lalla, Nicola Omodei, Niccolò Bucciantini, **Jack T. Dinsmore**, Nicolò Cibrario, and others. **X-Ray Polarization Detection of the Pulsar Wind Nebula in G21.5–0.9 with IXPE**. *ApJ*, 991(1):67, September 2025
- 8. Tobin M. Wainer, Gail Zasowski, Joshua Pepper, Tom Wagg, Christina L. Hedges, and others. **Catalog of Integrated-Light Star Cluster Light Curves in TESS**. *AJ*, 166(3):106, August 2023
- 9. Josephine Wong, Roger W. Romani, and **Jack T. Dinsmore**. **Improved Measurements of the IXPE Crab Polarization**. *ApJ*, 953(1):28, July 2023

- 10. Jeffrey Krupa, Kelvin Lin, Maria Acosta Flechas, **Jack Dinsmore**, Javier Duarte, and others. [GPU Coprocessors as a Service for Deep Learning Inference in High Energy Physics](#). *Mach. Learn.: Sci. Technol.*, 2(3):035005, April 2021
- 11. **Jack Dinsmore**, Patrick Draper, David Kastor, Yue Qiu, and Jennie Traschen. [Schottky Anomaly of deSitter Black Holes](#). *Class. Quant. Grav.*, 37(5):054001, 2020

CONTRIBUTING AUTHOR NON-REFEREED PUBLICATIONS

- 0. Christopher Layden, Kevin Burdge, John J. Piotrowski, Gábor Furesz, Juliana García-Mejía, and others. [The Light-speed Project: High-Speed, Ultra-Low Read Noise Imaging and Polarimetry for the Magellan Telescopes](#). In *Ground-based and Airborne Instrumentation for Astronomy XI*, volume 14149, page 141492I. International Society for Optics and Photonics, SPIE, 2026